



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St.,Lagao, General Santos City

Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 1 of 2
Document Code: WI-02	EVALUATION OF MEASUREMENT UNCERTAINTY		

Introduction

This document gives the procedure for the estimation of measurement uncertainty (MU) and applies to quantitative methods used in the BFAR RFL XII.

1. Responsibilities

1. Technical Manager
2. Quality Assurance Manager
3. Laboratory Analysts

2. Procedure

- 2.1. Identify and evaluate all contributors to the measurement uncertainty, such as steps in the procedure or equipment.
- 2.2. Tabulate the components of measurement uncertainty in LF W02-CHE-01(List of Contributory Factors for Measurement of Uncertainty in Chemistry) in the form of standard deviations.
- 2.3. Construct the fishbone diagram.
- 2.4. Calculate the combined uncertainty of each contributing equipment or step, followed by the overall combined uncertainty using representative data.
- 2.5. Calculate the expanded uncertainty of the method.
- 2.6. Check all the calculations.
- 2.7. File the list of contributors to the MU, the fishbone diagrams, and the calculations in LF W02-CHE-02 (Estimation of Measurement Uncertainty for Histamine).

3. Forms

LF W02-CHE-01, List of Contributory Factors for Measurement of Uncertainty in Chemistry

LF W02-CHE-02, Estimation of Measurement Uncertainty for Histamine Data Sheet

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ANNEX A: FORMULAE

Relative Standard Deviation, RSD

$$= \frac{\text{Standard Deviation}}{\text{Mean}} \times 100\%$$

Combined Uncertainty (u_c)

$$= \text{SQRT} [u_{(x)}^2 + u_{(y)}^2 + u_{(z)}^2 \dots]$$

where SQRT denotes the square root of the quantity in brackets and $u_{(x)}$ and so on indicate the individual uncertainties (in the form of standard deviations) that contribute to the combined uncertainty of a particular step or equipment

Combined Uncertainty of the method, u_c/c_x

$$= \text{SQRT} [(u_{c1}/c_1) + (u_{c2}/c_2) + (u_{c3}/c_3) + \dots]$$

where u_c denotes the combined uncertainty of the measurement obtained from the contributing step or equipment and c_x stands for the measurement value.

Expanded Uncertainty, u_{exp}

$$= (u_c/c_x) \times k$$

where k is an appropriate coverage factor (usually $k=2$, for a confidence level of 95%)

Reference: *EURACHEM/CITAC Guide for Quantifying Uncertainty in Analytical Measurement*, 3rd edition. EURACHEM/CITAC. 2012

6 DOCUMENT HISTORY

Revision No.	Issue Date	Original Entry	Revised as	Recall Date

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